SaaS Subscription Metrics Dashboard

An end-to-end Power BI analytics project modeling MRR, churn, retention, and secure regional access for a subscription business.

About this project:

This dashboard was built to practice real Data Analyst and Analytics Engineering skills using a subscription/billing dataset. Rather than a simple sales dashboard, the goal was to model the core metrics a real SaaS company tracks internally — recurring revenue, churn, and customer retention over time — while also implementing enterprise-level features like row-level security. It's based on the IBM Telco Customer Churn dataset, with signup dates, churn dates, and region derived to enable time-based and cohort analysis.

Questions this dashboard answers:

What is our current MRR, and how has it changed month over month?
What's our churn rate, and is it improving or worsening?
How well do we retain customers over time?
Which contract type, region, or segment is most at risk?
What's the estimated customer lifetime value?

Skills demonstrated:

Power Query (ETL & data cleaning)
• Data Modeling (star schema, dual-date relationships)
• DAX (time intelligence, ALLEXCEPT, USERELATIONSHIP)
• Row-Level Security (region-based access control) •
Cohort Retention Analysis
• Custom Theming
• Published Live Report

SaaS Revenue & Customer Health

Executive overview of recurring revenue, customer activity and churn performance

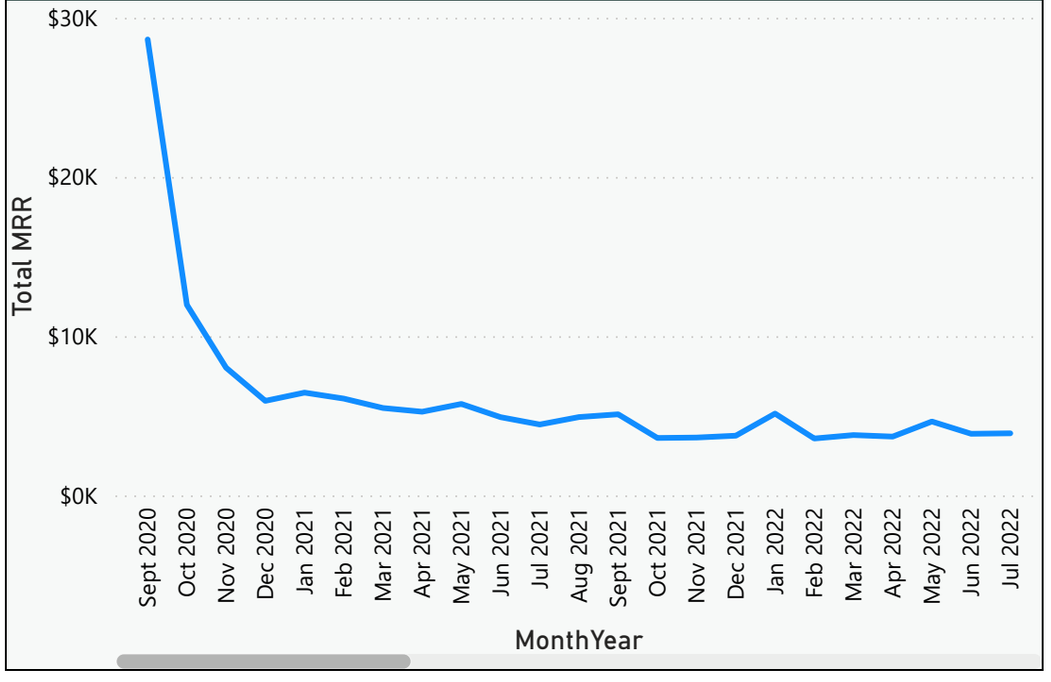
Monthly Recurring Revenue	ACTIVE CUSTOMERS	CHURN RATE	Average Revenue Per User
$316.99K	5.17K	26.54%	$61.27

Region: All

Contract Type: All

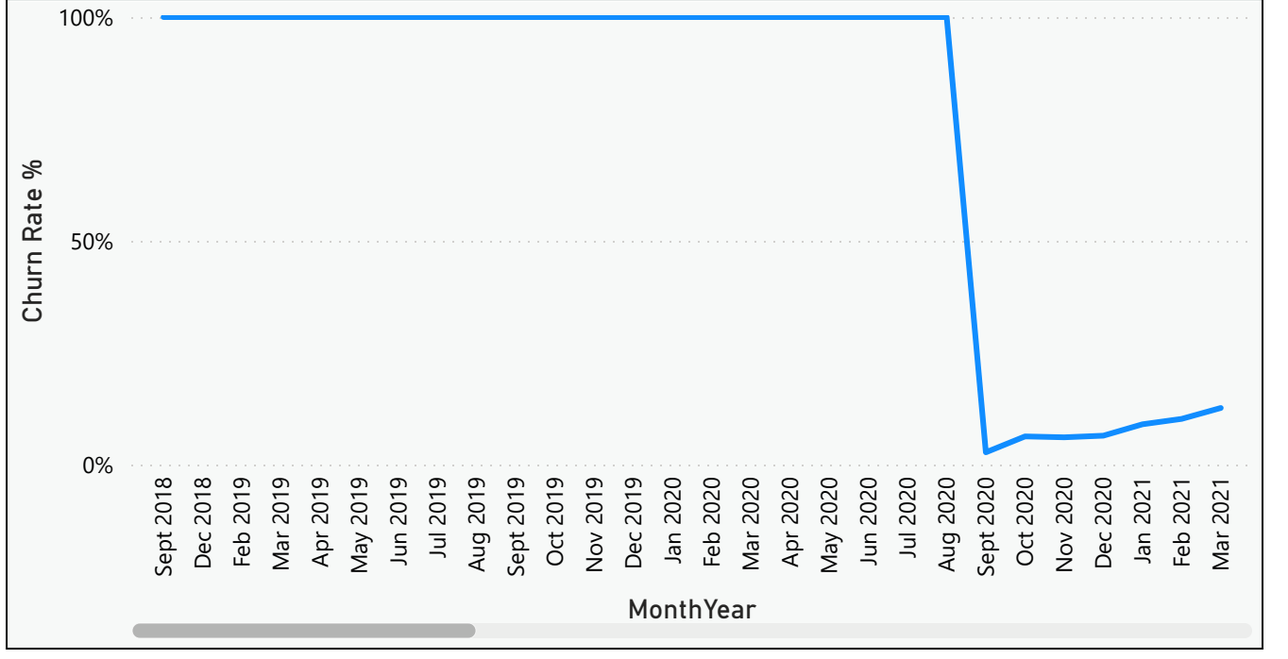
Monthly Recurring Revenue

MRR trend over time



Customer Churn Rate

Monthly churn trend



Key Insight: MRR by month reflects revenue from each signup cohort rather than a cumulative running total. Churn rate appears artificially high in the earliest months (2018-2019) due to very small customer counts in that period — the trend from late 2020 onward, once volume stabilizes, is the more reliable read.

Customer Retention & Cohort Analysis

Understanding how customer retention changes across signup cohorts

AVG COHORT RETENTION	BEST COHORT RETENTION	LOWEST COHORT RETENTION	TOTAL COHORTS
74.3%	100.0% Aug 2026	42.2% Dec 2024	93

CUSTOMER RETENTION BY COHORT

Cohort Month	Total Customers	Active Customers	Cohort Retention %
Apr 2019	5		
Apr 2020	10		
Apr 2021	73	67.00	91.8%
Apr 2022	65	56.00	86.2%
Apr 2023	80	56.00	70.0%
Apr 2024	93	57.00	61.3%
Apr 2025	105	61.00	58.1%
Apr 2026	124	93.00	75.0%
Aug 2019	3		
Aug 2020	10		
Aug 2021	77	68.00	88.3%
Aug 2022	67	51.00	76.1%
Aug 2023	67	50.00	74.6%
Aug 2024	120	56.00	46.7%
Aug 2025	104	71.00	68.3%
Aug 2026	11	11.00	100.0%
Dec 2018	1		
Dec 2019	6		
Dec 2020	93	87.00	93.5%
Dec 2021	70	57.00	81.4%
Total	**7043**	**5,174.00**	**73.5%**

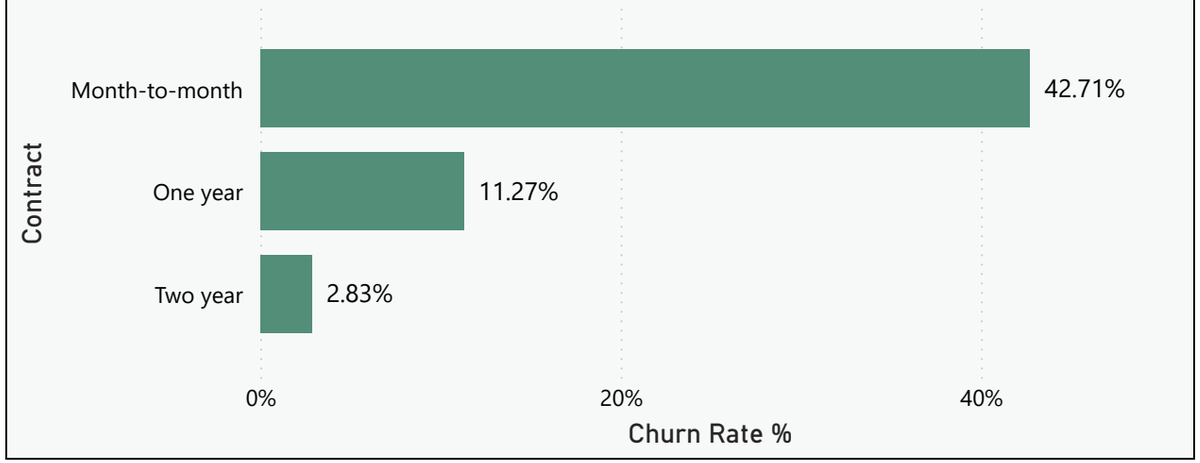
Key Insight: Retention declines as customer tenure increases — customers who stay past their first year are significantly more likely to remain long-term, while the highest drop-off happens early in the customer lifecycle. Average cohort retention is 74.3%. Early cohorts (2018-2020) show blank or extreme values due to very small sample sizes (as few as 1-10 customers) and shouldn't be read as meaningful trends — cohorts from 2021 onward, with 60-120+ customers each, give the reliable signal.

Churn & Customer Risk Analysis

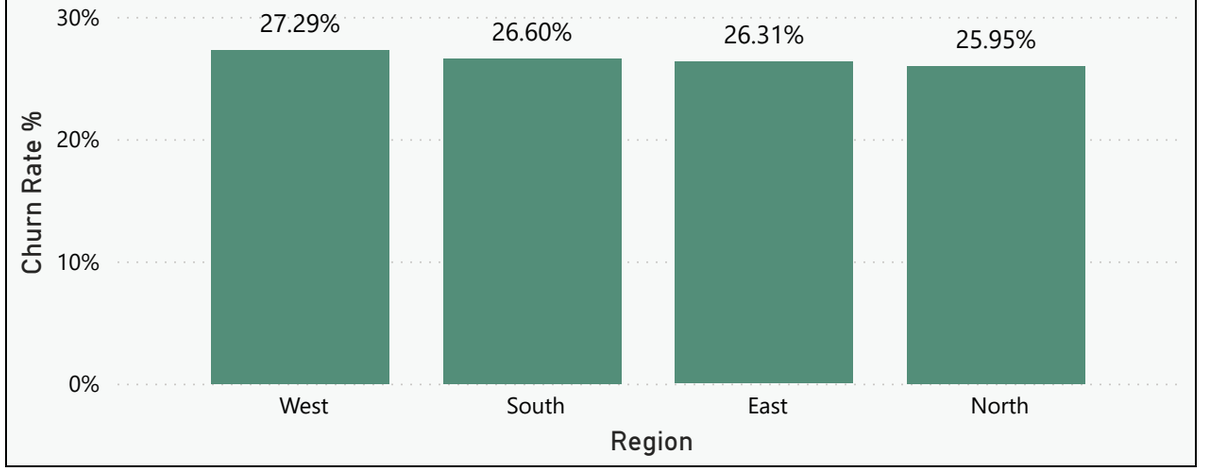
Identify customer segments and accounts contributing to subscription churn

CHURNED RATE	RETENTION RATE	CHURNED MRR	CHURNED CUSTOMERS
26.54%	73.46%	$139.13K	2K

Churn Rate by Contract



Churn Rate by Region



HIGH-RISK CUSTOMER ACCOUNTS

CustomerID	Sum of Monthly Charges	Sum of Tenure Months	Contract	Risk Score
5709-LVOEQ	80.85	0	Two year	80.85
4075-WKNIU	73.35	0	Two year	73.35
2775-SEFEE	61.90	0	Two year	61.90
1371-DWPAZ	56.05	0	Two year	56.05
4472-LVYGI	52.55	0	Two year	52.55
1393-IMKZG	95.85	1	Month-to-month	47.93
1941-HOSAM	90.10	1	Month-to-month	45.05
4273-MBHYA	89.35	1	Month-to-month	44.68
Total	**316,985.75**	**194387**		**8,218.46**

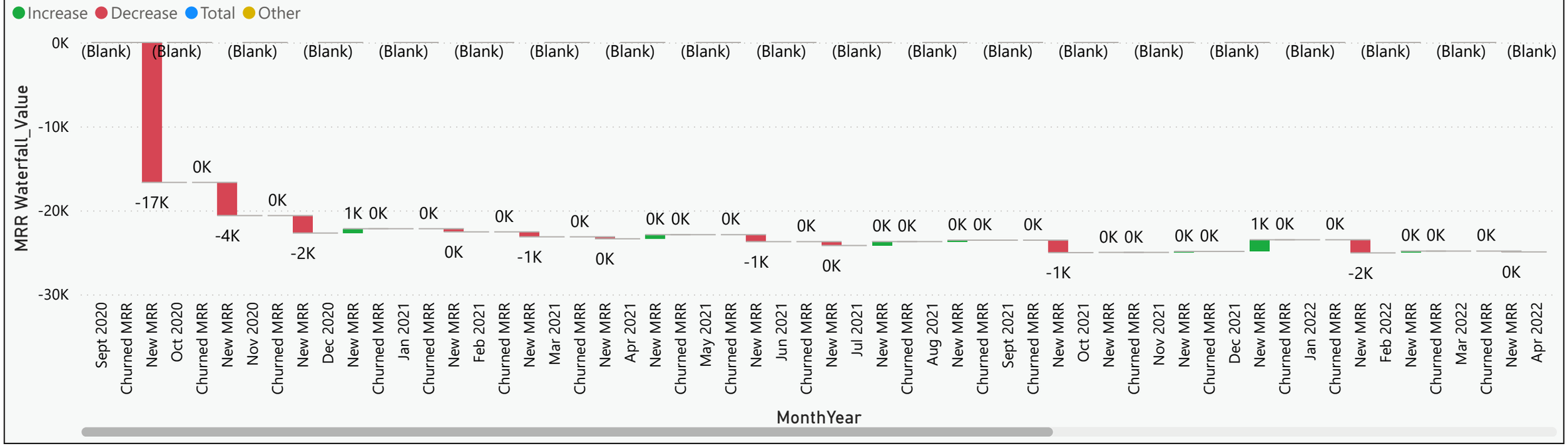
Key Insight: Contract type is the strongest churn driver by far — month-to-month customers churn at 42.71%, nearly 15x the rate of two-year contract customers (2.83%). Regional churn variation (26-27%) is minor and likely reflects this dataset's synthetic region assignment rather than a true business pattern.

MRR Movement & Revenue Performance

Monthly recurring revenue movement across the reporting period

MRR CHANGE	MRR CHANGE	AVERAGE MRR	ACTIVE CUSTOMERS
($8.38K)	-0.95	$4.40K	5.17K

MRR Movement: New vs. Churned Revenue



Key Insight: Churned MRR only appears from mid-2024 onward due to how churn dates were derived in this dataset (churn events were modeled within the last 24 months). Prior to that period, the chart reflects new MRR additions only. From 2024 onward, the waterfall shows genuine month-to-month tension between new signups and churned revenue.